COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 12 : SLO – 1 | Logic gates (combined)
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I. Match the Following

• 1. Symbol of AND gate — standard AND symbol (■), Y = A■B
• 2. Output of OR gate when A=0, B=1 — 1
• 3. NOT gate logic function — c) Y = A■
• 4. NOR gate implementation — d) Universal gate to implement AND, OR, NOT
• 5. EX-OR practical use — e) Parity generator/checker, Half Adder

II. Fill in the Blanks

• 1. all 2. any 3. 0 4. universal 5. different

III. Activity / Describe and Perform

• 1. AND/OR/NOT — AND: Y=A■B, truth table 00→0, 01→0, 10→0, 11→1. OR: Y=A+B, 00→0, 01→1, 10→1, 11→1. NOT: Y=A■, 0→1, 1→0. Symbols: D-shape (AND), curved (OR), triangle+bubble (NOT).
• 2. NOR gate: Y = (A+B)■ — output 1 only when both inputs are 0. Truth table: 00→1, 01→0, 10→0, 11→0. Implementations: NOT = A NOR A; OR = (A NOR B) NOR (A NOR B); AND = (A NOR A) NOR (B NOR B).
• 3. EX-OR gate: Y = A⊕B = A■B + AB■ — output 1 when inputs differ. Truth table: 00→0, 01→1, 10→1, 11→0. Uses: parity generator/checker, Half Adder sum output, digital comparators.